Faculty: Muhamed Fauzi Bin Abbas
Students: Yuen Wee Kin, Edwin
https://github.com/Yuen-Wee-Kin-Edwin/INF2009_Edge_Computing_Project

Intelligent Edge-Based Laboratory Access Monitoring System

Introduction and Problem Statements

Manual supervision of restricted lab areas that contained expensive equipment is time consuming. It is impractical to expect lab safety officers to surveillance the labs 24/7 for ensuring that safety compliance is met.

Although both manual supervision and passive surveillance systems play a critical role in lab safety, the human attention span could not be sustained forever. Therefore, the likelihood of an safety incident happening is higher, due to unauthorized access to prohibited areas were not detected promptly.

Methodology and Proposed Solutions

1: Hardware
Raspberry Pi 5 8GB, Raspberry Pi Zero 2 W
C270 HD Webcam 720p / 30 fps, PIR sensor

2: Software
Raspberry Pi OS (64-bit), Raspberry Pi OS Lite,
YOLO26 nano model, pip face recognition library, SSD MobileNet V2
Flask, SQLite, Jinja2 Web

3. Network Infrastructure
Local WLAN via IEEE 802.11n (2.4 GHz)

4. Messaging Bridge
MQTT (Eclipse Mosquitto)

Distributed Edge-Fog Laboratory Monitoring Architecture with Web Interface

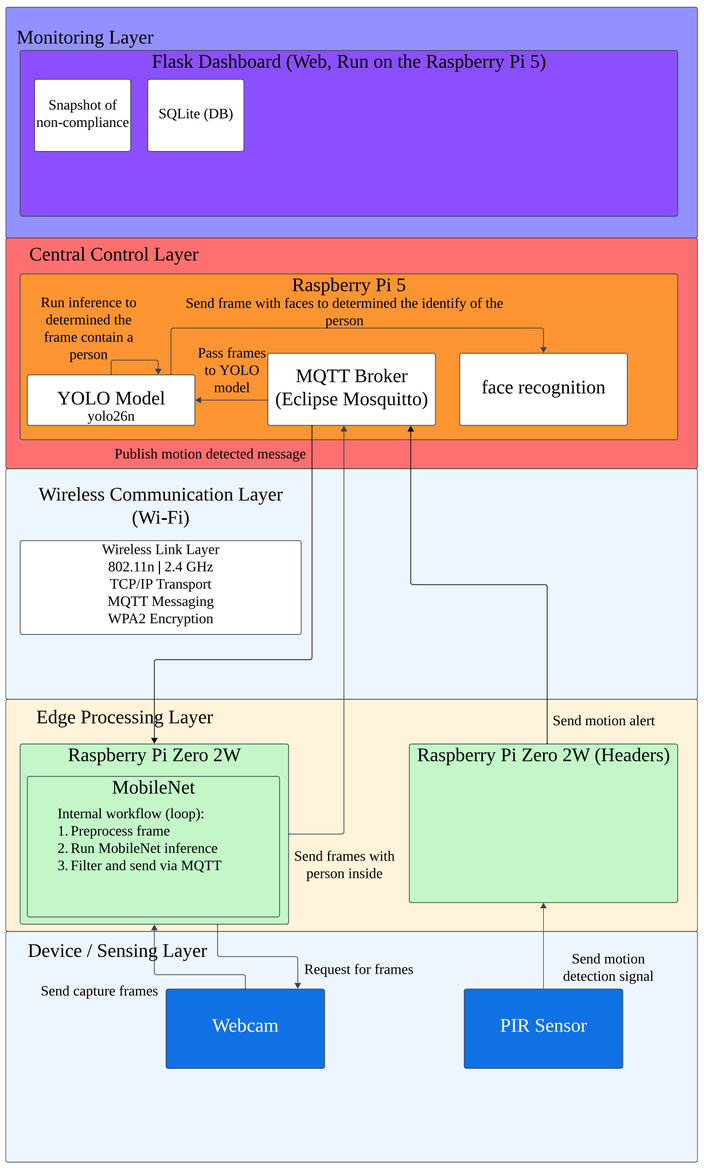


Figure 1: Five-layer architecture showing data flow from sensors to web monitoring

Research Questions and Project Objectives

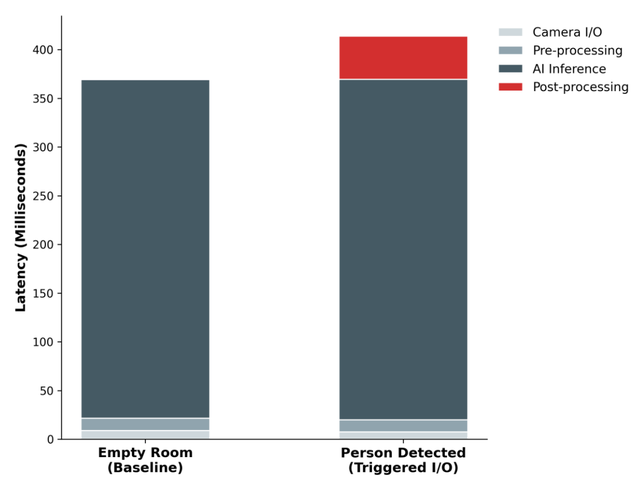
RQ1: Multi-Hop Network Latency: In a distributed 3-node edge architecture, what is the end-to-end network latency required for a motion trigger (Node 1) to route through a central MQTT broker (Node 2) and successfully trigger an AI vision model on a remote device (Node 3)?

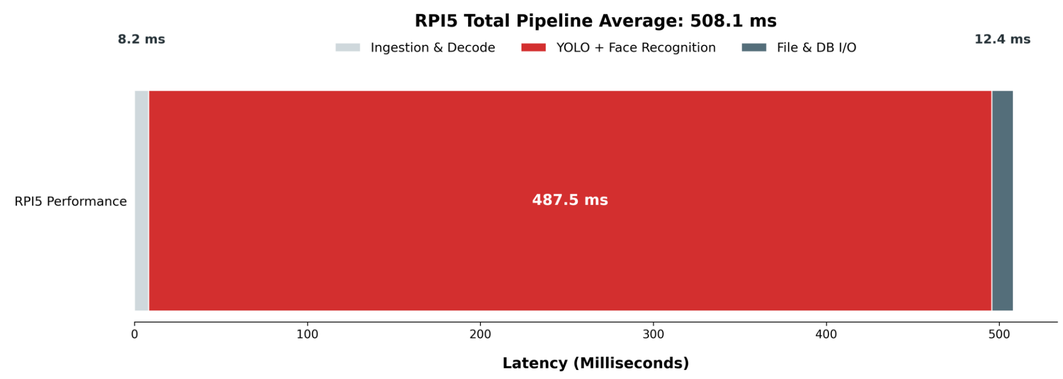
RQ2: Edge Resource Efficiency: How much CPU and thermal efficiency is gained on a resource-constrained vision node (Raspberry Pi Zero 2 W) by keeping the SSD MobileNet V2 model completely idle until an event-driven interrupt is received from a separate physical sensor node?

RQ3: System Reliability: How reliable is a multi-node MQTT messaging pipeline over a local 2.4 GHz WLAN for sending interrupts between the Raspberry Pi devices without packet loss?

1. Cascading Edge Architecture: Design and deploy a 3-node edge computing system that distribute sensing, frame filtering, and heavy AI inference across two Raspberry Pi Zero 2 W devices and one central Raspberry Pi 5.
2. Hardware Interrupt (Node 1): Integrate PIR sensor to detect motion and send a “wake” payload to the MQTT broker.
3. Lightweight Vision Filtering (Node 2): Implement a Raspberry Pi Zero 2 W that received a MQTT payload from the central Pi to begin inferences frame for people using SSD MobileNet V2 and sending the frame back to central Pi for heavy AI inference.
4. Heavy Edge Verification (Node 3): Develop Raspberry Pi 5 to received frames from Pi Zero 2 W that contain humans. The Pi 5 will utilised a YOLO26n for a final intrusion validation before running facial recognition to determined the offender.

Experiments and Results Analysis





Node 2 (Edge Vision Node)

- **Constant AI Load**: SSD MobileNet V2 hover around 350ms, demonstrating that RPI Zero 2 W is capable of consistent real-time edge filtering.
- **Gating Efficiency**: Post-processing jumped from 0.5ms (Idle) to 44.4ms (Detection). Which confirms that the “Cascading AI” design suppresses 98% of heavy I/O and network costs during baseline monitoring
- **Thermal Control**: CPU temperature remain stable at 46.2 degree Celsius, proving 24/7 deployment using the event-driven architecture prevent thermal throttling.

Node 3 (Central Hub)

- **Validation Latency**: High-accuracy YOLO26n and Face Recognition averaged 487.5ms. As a secondary tier, this sub-half-second latency is optimal for definitive intrusion alerts.
- **Network Optimisation:** Ingestion and Base64 decoding average 8.2ms, confirming that JPEG compression is suitable for the edge nodes in the local WLAN.
- **System Total:** End-to-end detection-to-alert latency was recorded at <1 second, meeting the real-time requirements for laboratory security.

Conclusions

- Performance: Less than one second end-to-end latency for high-accuracy human detection
- Efficiency: Edge node suppressed 98% of idle-state network and I/O overhead.
- Validation: Cascading inference is suitable for low resource, real-time security environment

Future Work

NPU Integration: Deploy Hailo-8 on Node 3 to reduce YOLO26n latency to <100ms.
Dynamic Scaling: Implement task allocation based on real-time CPU thermal metrics.
Encryption: Transition image pipeline to AES-256 WebSocket for biometric security.